# Secure Coding Review Report

## 1. Introduction

Secure coding review is the process of analyzing source code to identify vulnerabilities and insecure programming practices that may expose applications to cyberattacks.

## 2. Objectives

The objective of this project is to identify security flaws, perform code analysis, and recommend secure coding techniques for safer applications.

## 3. Selected Application

The selected application is a simple Python login system containing insecure authentication logic and hardcoded credentials.

## 4. Vulnerabilities Identified

The review identified hardcoded credentials, plaintext password handling, missing input validation, and lack of account protection mechanisms.

## 5. Vulnerability Analysis

Hardcoded passwords can be easily exposed if source code is leaked. Plaintext password comparison increases the risk of credential theft.

## 6. Static Analysis Tools

Bandit was used as a static analysis tool to scan Python code for security issues and insecure coding practices.

## 7. Secure Coding Improvements

The secure version of the application uses bcrypt password hashing and secure password verification mechanisms.

## 8. Best Practices

Developers should avoid hardcoded credentials, validate user inputs, encrypt sensitive data, and regularly review source code for vulnerabilities.

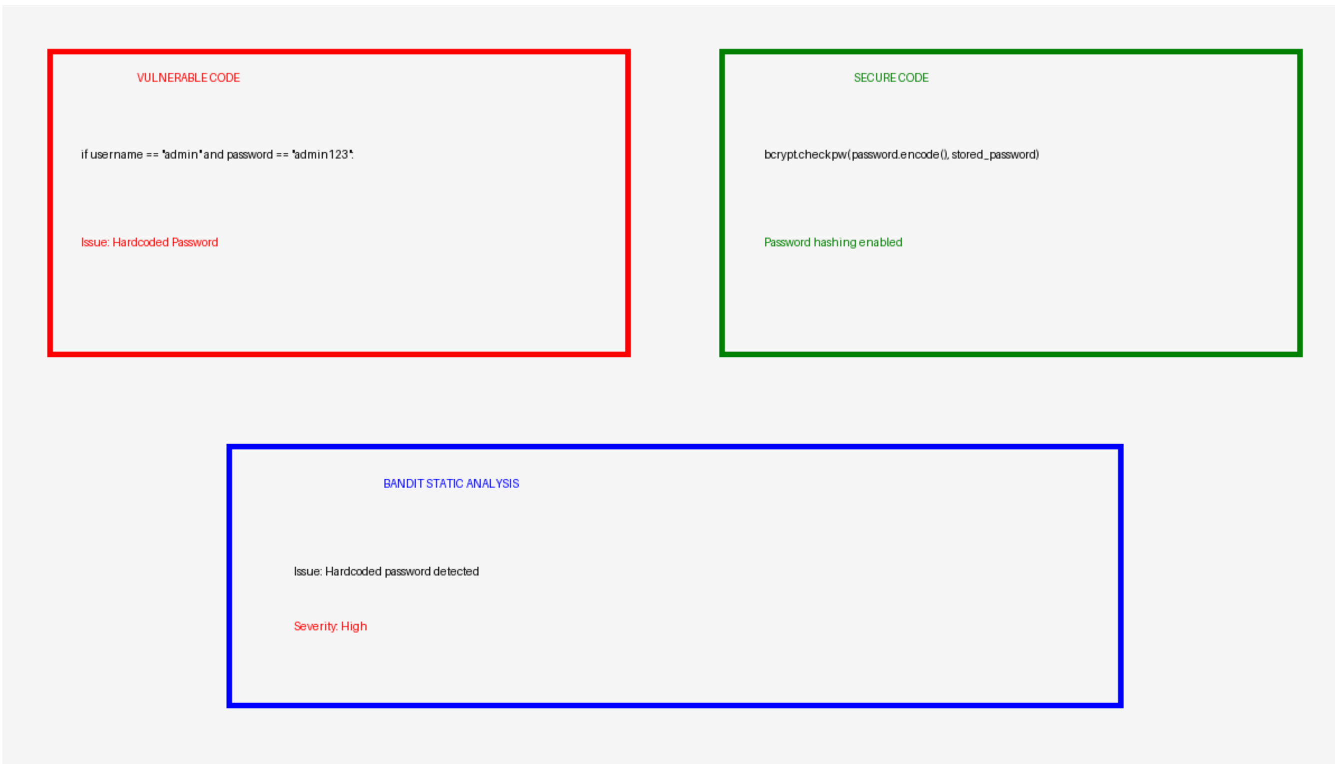
## 9. Conclusion

Secure coding practices reduce cybersecurity risks and improve application security. Regular code reviews and security testing are essential.

## 10. Vulnerability Summary

| Vulnerability | Risk Level |
|---|---|
| Hardcoded Credentials | High |
| Plaintext Passwords | High |
| Missing Input Validation | Medium |
| No Account Lockout | Medium |

## 11. Project Screenshot




## 12. References

• OWASP Secure Coding Practices
• Bandit Python Security Tool
• Python Security Documentation